CPU 流水线验证报告

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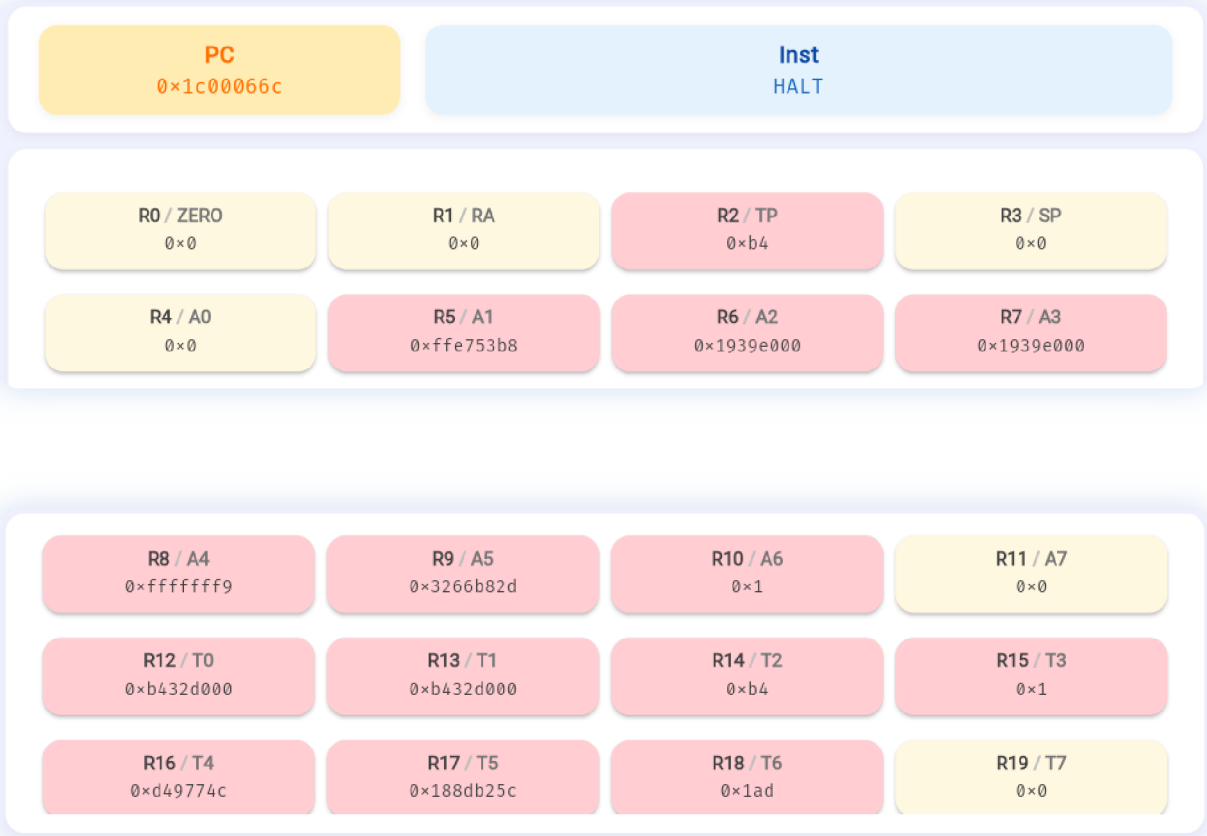
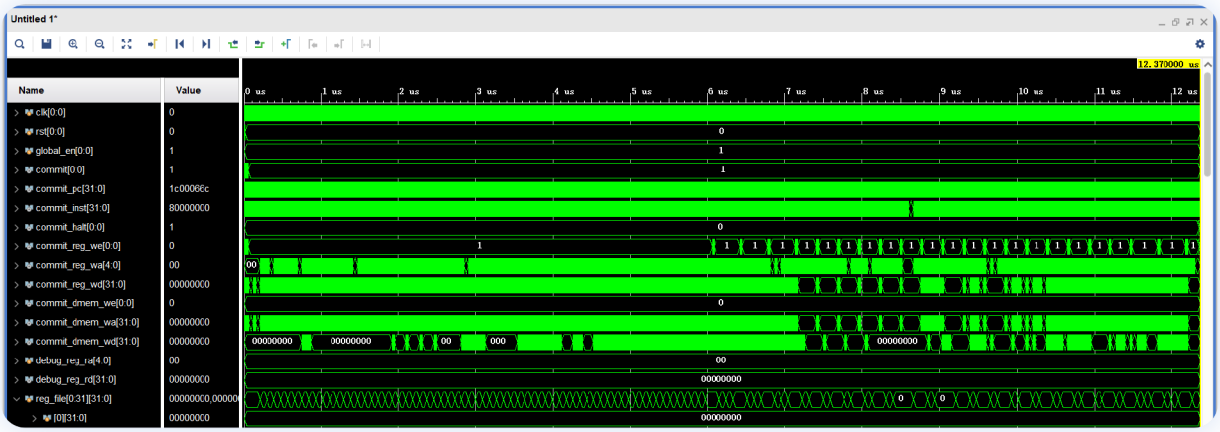
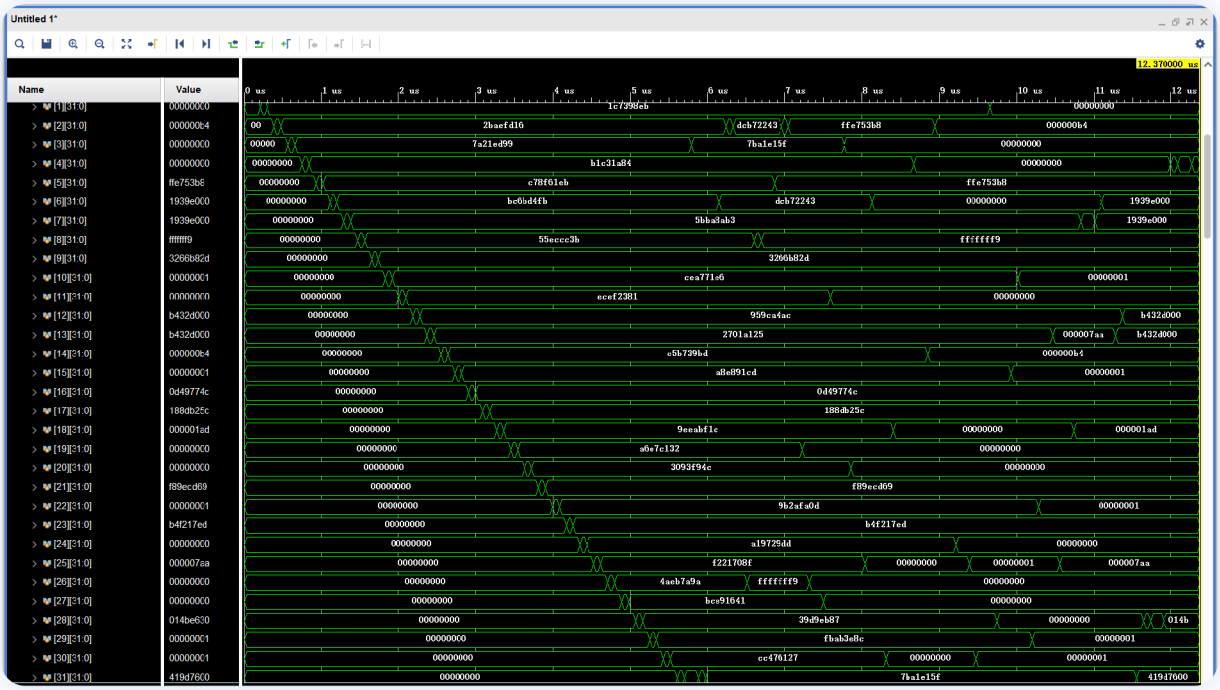
测试 1：无冒险流水线

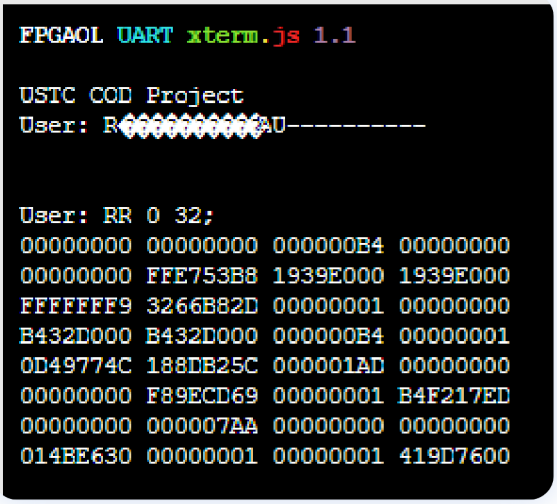
经比对，理论结果、仿真结果、上板结果三者一致。



R20 / T8 0×0	R21 0×f89ecd69	R22 / S9 / FP 0×1	R23 / S0 0×b4f217ed
R24 / S1 0×0	R25 / S2 0×7aa	R26 / S3 0×0	R27 / S4 0×0
R28 / S5 0×14be630	R29 / S6 0×1	R30 / S7 0×1	R31 / S8 0×419d7600

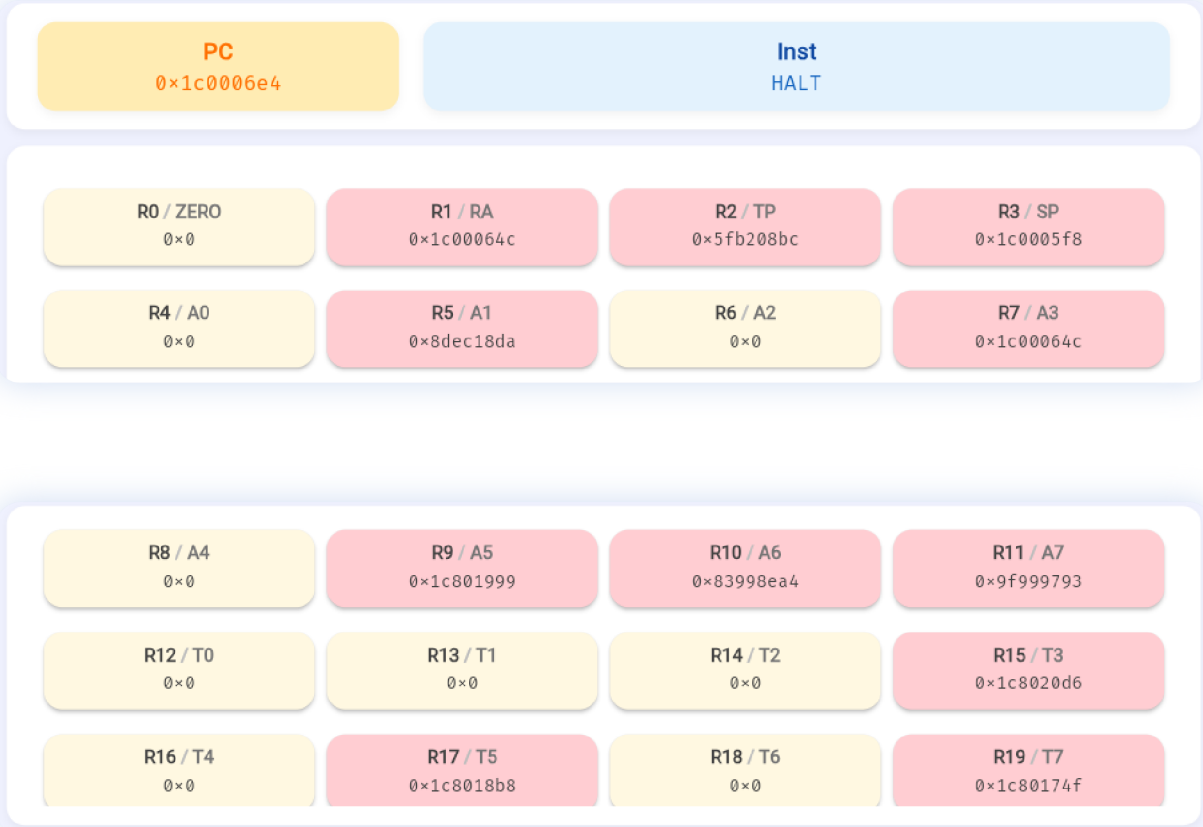




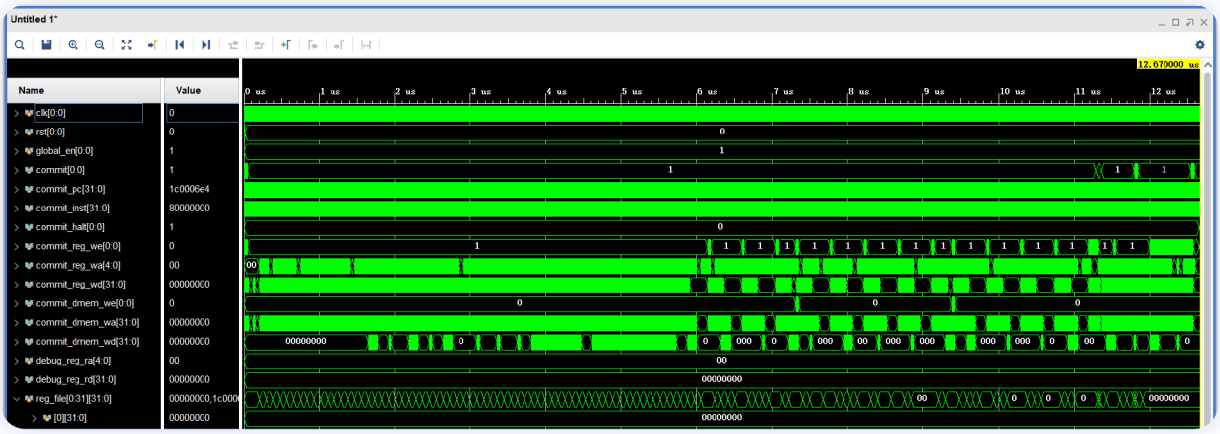


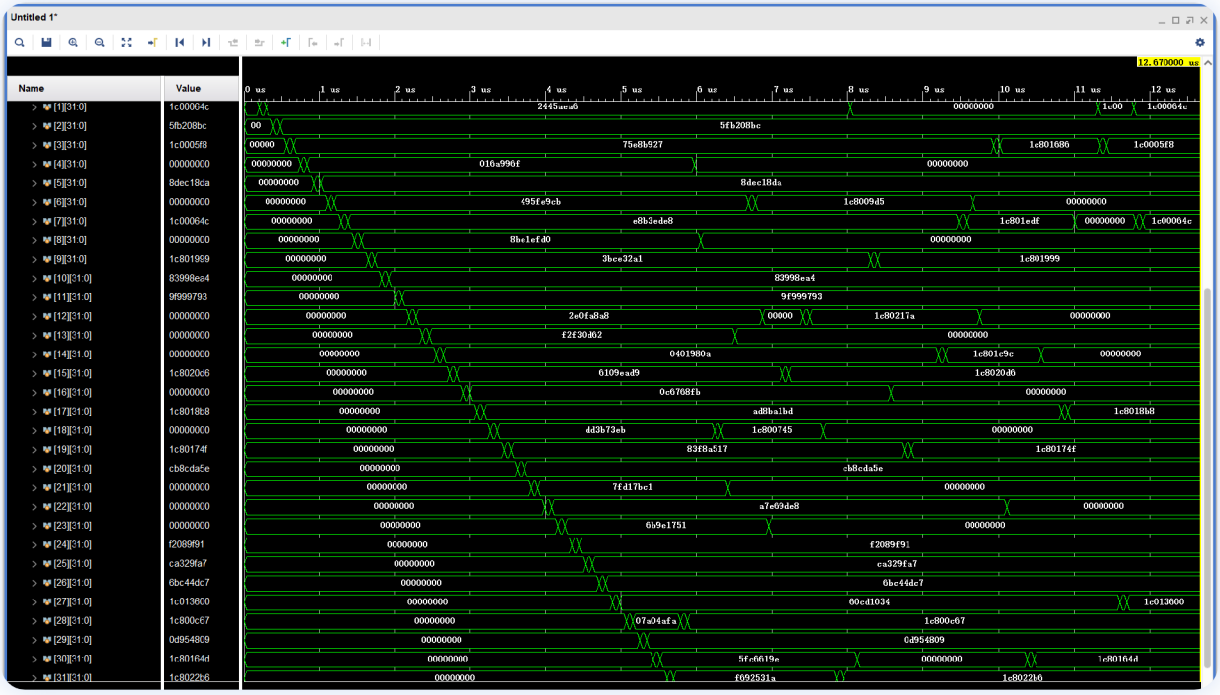
测试 2：无冒险流水线

经比对，理论结果、仿真结果、上板结果三者一致。



R20 / T8 0×cb8cda5e	R21 0×0	R22 / S9 / FP 0×0	R23 / S0 0×0
R24 / S1 0×f2089f91	R25 / S2 0×ca329fa7	R26 / S3 0×6bc44dc7	R27 / S4 0×1c013600
R28 / S5 0×1c800c67	R29 / S6 0×d954809	R30 / S7 0×1c80164d	R31 / S8 0×1c8022b6





```
FPGAOL UART xterm.js 1.1

USTC COD Project
User: R◆◆◆◆◆◆◆◆◆◆◆◆AU----------


User: RR 0 32;
00000000 1C00064C 5FB208BC 1C0005F8
00000000 8DEC18DA 00000000 1C00064C
00000000 1C801999 83998EA4 9F999793
00000000 00000000 00000000 1C8020D6
00000000 1C8018B8 00000000 1C80174F
CB8CDA5E 00000000 00000000 00000000
F2089F91 CA329FA7 6BC44DC7 1C013600
1C800C67 0D954809 1C80164D 1C8022B6
```

测试 3：加入前递的流水线

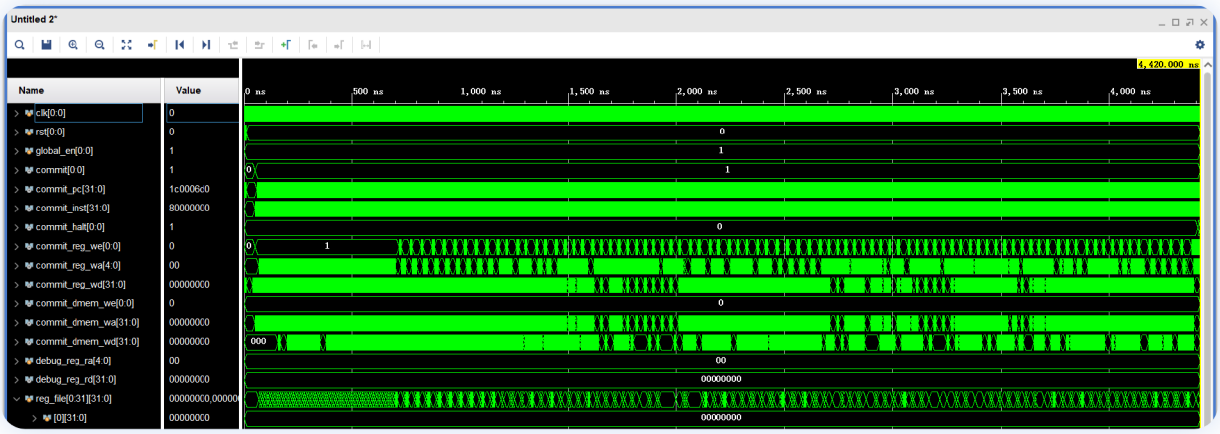
经比对，理论结果、仿真结果、上板结果三者一致。

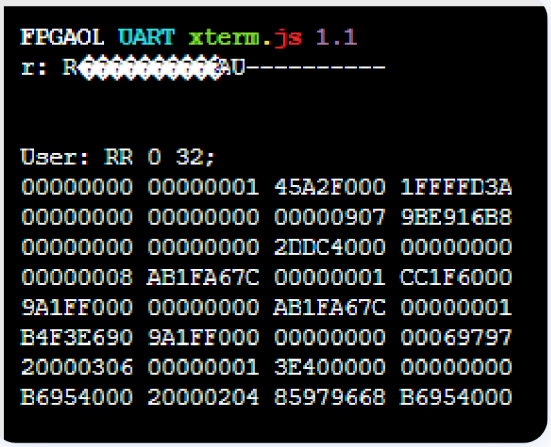
PC	Inst
0×1c0006d0	HALT

R0 / ZERO 0×0	R1 / RA 0×1	R2 / TP 0×45a2f000	R3 / SP 0×1ffffd3a
R4 / A0 0×0	R5 / A1 0×0	R6 / A2 0×907	R7 / A3 0×9be916b8

R8 / A4 0×0	R9 / A5 0×0	R10 / A6 0×2ddc4000	R11 / A7 0×0
R12 / T0 0×8	R13 / T1 0×ab1fa67c	R14 / T2 0×1	R15 / T3 0×cc1f6000
R16 / T4 0×9a1ff000	R17 / T5 0×0	R18 / T6 0×ab1fa67c	R19 / T7 0×1

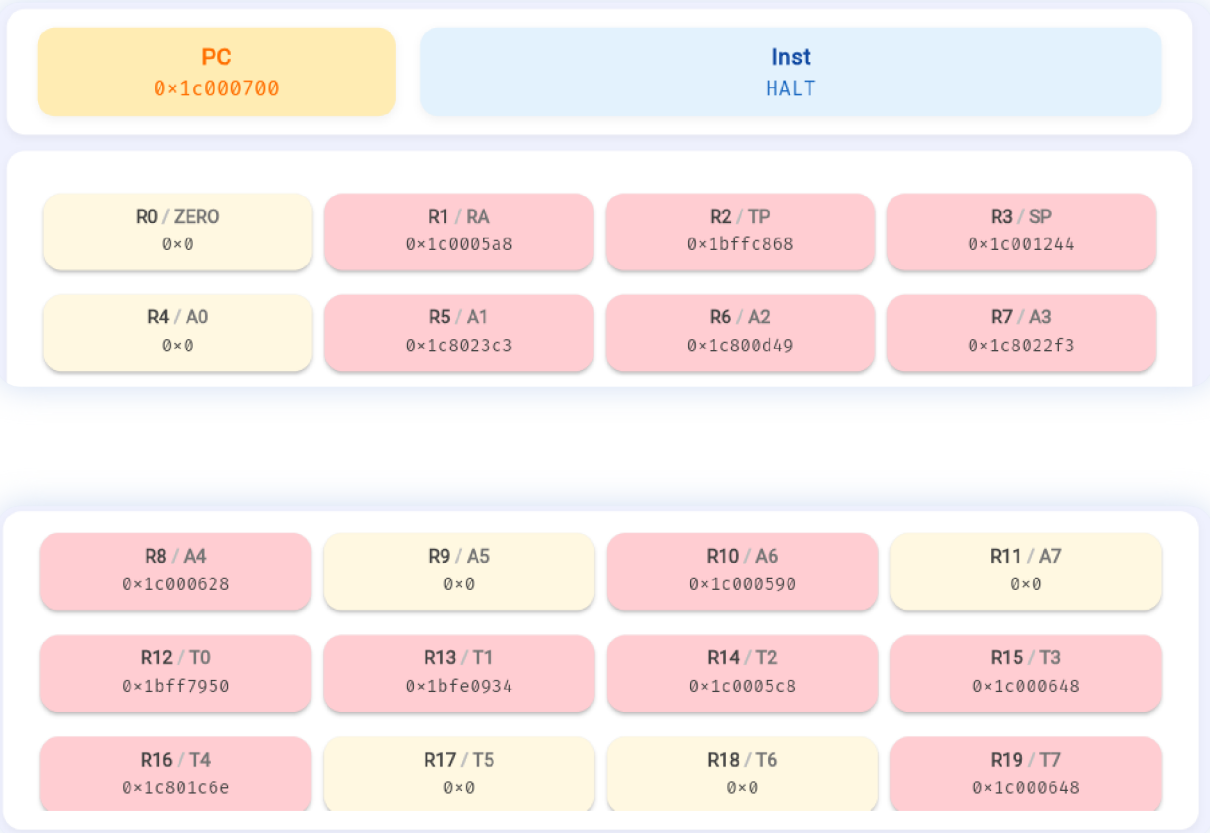
R20 / T8 0×b4f3e690	R21 0×9a1ff000	R22 / S9 / FP 0×0	R23 / S0 0×69797
R24 / S1 0×20000306	R25 / S2 0×1	R26 / S3 0×3e400000	R27 / S4 0×0
R28 / S5 0×b6954000	R29 / S6 0×20000204	R30 / S7 0×85979668	R31 / S8 0×b6954000



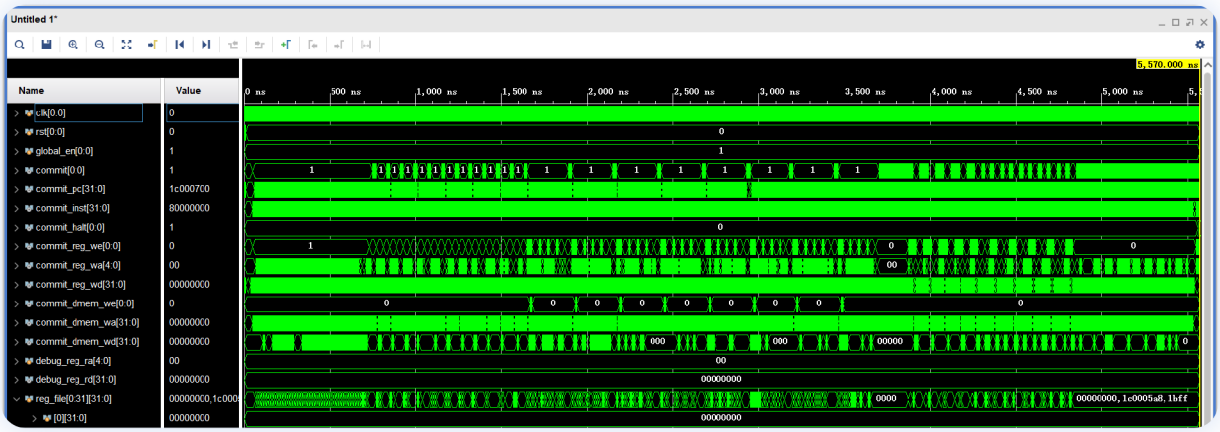


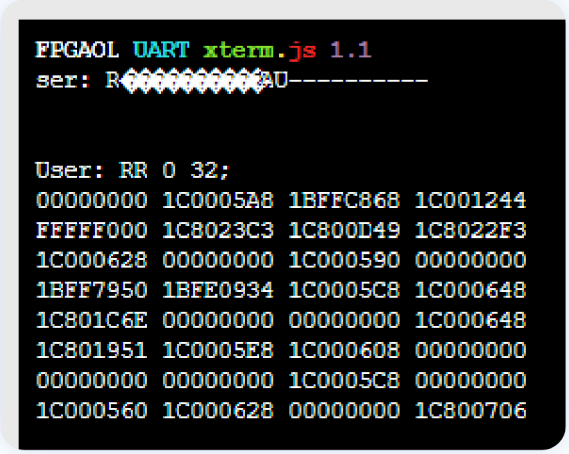
测试 4：完整流水线 CPU

经比对，理论结果、仿真结果、上板结果三者一致。(说明：上板结果中 R4 之所以是 FFFFF000，是因为测试程序 II 中的 win, lose label 间没有 nop 分隔，而 halt 之后流水线并未立即停止)

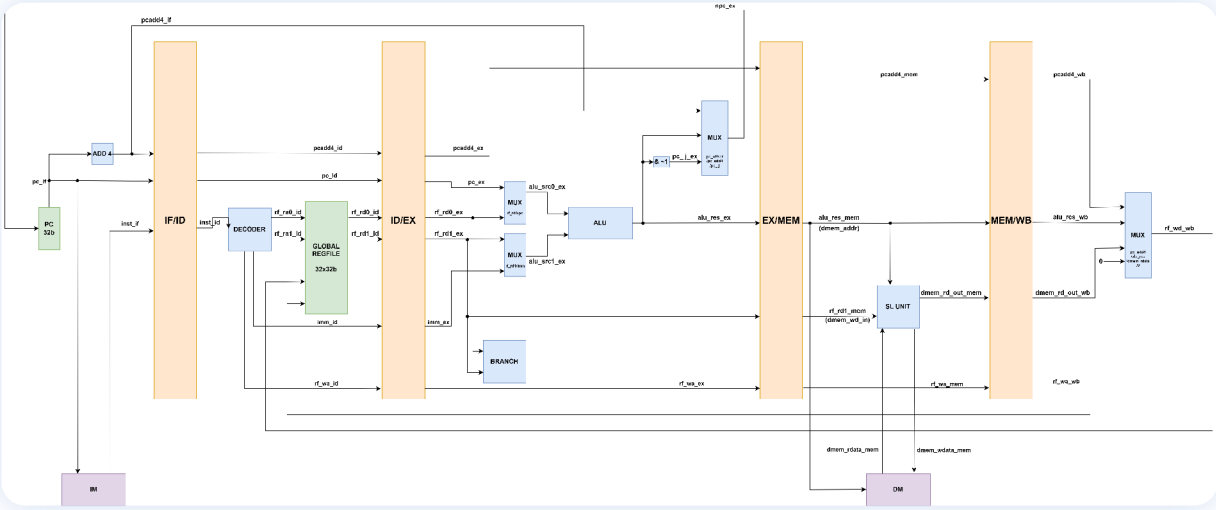


R20 / T8 0×1c801951	R21 0×1c0005e8	R22 / S9 / FP 0×1c000608	R23 / S0 0×0
R24 / S1 0×0	R25 / S2 0×0	R26 / S3 0×1c0005c8	R27 / S4 0×0
R28 / S5 0×1c000560	R29 / S6 0×1c000628	R30 / S7 0×0	R31 / S8 0×1c800706





附录 A：简单流水线 CPU 数据通路图



注：上图展示了五级流水线的基准数据通路。为了保持架构图的清晰易懂，此处省略了已在 RTL 中实现的前递路径与冒险检测逻辑。

附录 B：仿真文件与测试程序

Top

```
module Top (
    input                   [ 0 : 0]            clk,
    input                   [ 0 : 0]            rst,

    input                   [ 0 : 0]            global_en,

/* ---------------------------------- Debug ---------------------------------- */
    output                  [ 0 : 0]            commit,
    output                  [31 : 0]            commit_pc,
    output                  [31 : 0]            commit_inst,
    output                  [ 0 : 0]            commit_halt,
```

```
    output                  [ 0 : 0]            commit_reg_we,
    output                  [ 4 : 0]            commit_reg_wa,
    output                  [31 : 0]            commit_reg_wd,
    output                  [ 0 : 0]            commit_dmem_we,
    output                  [31 : 0]            commit_dmem_wa,
    output                  [31 : 0]            commit_dmem_wd,

    input                   [ 4 : 0]            debug_reg_ra,
    output                  [31 : 0]            debug_reg_rd
);

/* ----------------------------- Memory (inst) ----------------------------- */
wire                  [31 : 0]            imem_raddr;
wire                  [31 : 0]            imem_rdata;

/* ----------------------------- Memory (data) ----------------------------- */
wire                  [31 : 0]            dmem_rdata;
wire                  [ 0 : 0]            dmem_we;
wire                  [31 : 0]            dmem_addr;
wire                  [31 : 0]            dmem_wdata;

CPU cpu (
    .clk(clk),
    .rst(rst),
    .global_en(global_en),

    .imem_raddr(imem_raddr),
    .imem_rdata(imem_rdata),

    .dmem_rdata(dmem_rdata),
    .dmem_we(dmem_we),
    .dmem_addr(dmem_addr),
    .dmem_wdata(dmem_wdata),

    .commit(commit),
    .commit_pc(commit_pc),
    .commit_inst(commit_inst),
    .commit_halt(commit_halt),
    .commit_reg_we(commit_reg_we),
    .commit_reg_wa(commit_reg_wa),
    .commit_reg_wd(commit_reg_wd),
    .commit_dmem_we(commit_dmem_we),
    .commit_dmem_wa(commit_dmem_wa),
    .commit_dmem_wd(commit_dmem_wd),

    .debug_reg_ra(debug_reg_ra),
    .debug_reg_rd(debug_reg_rd)
);

INST_MEM inst_mem(
    .a(imem_raddr[10:2]),
    .d(0),
    .clk(clk),
    .we(0),
```

```
    .spo(imem_rdata)
);

DATA_MEM data_mem(
    .a(dmem_addr[10:2]),
    .d(dmem_wdata),
    .clk(clk),
    .we(dmem_we),
    .spo(dmem_rdata)
);
endmodule
```

Top_tb

```
module Top_tb();
reg                 [ 0 : 0]            clk;
reg                 [ 0 : 0]            rst;

reg                 [ 0 : 0]            global_en;

/* ---------------------------------- Debug ---------------------------------- */
wire                [ 0 : 0]            commit;
wire                [31 : 0]            commit_pc;
wire                [31 : 0]            commit_inst;
wire                [ 0 : 0]            commit_halt;
wire                [ 0 : 0]            commit_reg_we;
wire                [ 4 : 0]            commit_reg_wa;
wire                [31 : 0]            commit_reg_wd;
wire                [ 0 : 0]            commit_dmem_we;
wire                [31 : 0]            commit_dmem_wa;
wire                [31 : 0]            commit_dmem_wd;

reg                 [ 4 : 0]            debug_reg_ra;
wire                [31 : 0]            debug_reg_rd;

Top top (
    .clk(clk),
    .rst(rst),
    .global_en(global_en),

    .commit(commit),
    .commit_pc(commit_pc),
    .commit_inst(commit_inst),
    .commit_halt(commit_halt),
    .commit_reg_we(commit_reg_we),
    .commit_reg_wa(commit_reg_wa),
    .commit_reg_wd(commit_reg_wd),
    .commit_dmem_we(commit_dmem_we),
    .commit_dmem_wa(commit_dmem_wa),
    .commit_dmem_wd(commit_dmem_wd),

    .debug_reg_ra(debug_reg_ra),
```

```
    .debug_reg_rd(debug_reg_rd)
);

always #5 clk = ~clk;
initial begin
    clk = 1;
    debug_reg_ra = 0;
    global_en = 1;
    rst = 1;
    #10 rst = 0;
    forever begin
        #10;
        if (commit_halt == 1) begin
            $finish;
        end
    end
end
endmodule
```

测试程序 I

```
L0: # Randomly initialize GPR
    lu12i.w     $r0, 0xB3E26
    nop
    nop
    ori         $r0, $r0, 0xC50
    nop
    nop
    lu12i.w     $r1, 0x1C739
    nop
    nop
    ori         $r1, $r1, 0x8EB
    nop
    nop
    lu12i.w     $r2, 0x2BAEF
    nop
    nop
    ori         $r2, $r2, 0xD16
    nop
    nop
    lu12i.w     $r3, 0x7A21E
    nop
    nop
    ori         $r3, $r3, 0xD99
    nop
    nop
    lu12i.w     $r4, 0xB1C31
    nop
    nop
    ori         $r4, $r4, 0xA84
    nop
    nop
```

```
lu12i.w     $r5, 0xC78F6
nop
nop
ori         $r5, $r5, 0x1EB
nop
nop
lu12i.w     $r6, 0xBC6BD
nop
nop
ori         $r6, $r6, 0x4FB
nop
nop
lu12i.w     $r7, 0x5BBA8
nop
nop
ori         $r7, $r7, 0xAB3
nop
nop
lu12i.w     $r8, 0x55ECC
nop
nop
ori         $r8, $r8, 0xC3B
nop
nop
lu12i.w     $r9, 0x3266B
nop
nop
ori         $r9, $r9, 0x82D
nop
nop
lu12i.w     $r10, 0xCEA77
nop
nop
ori         $r10, $r10, 0x1E6
nop
nop
lu12i.w     $r11, 0xECEF2
nop
nop
ori         $r11, $r11, 0x381
nop
nop
lu12i.w     $r12, 0x959CA
nop
nop
ori         $r12, $r12, 0x4AC
nop
nop
lu12i.w     $r13, 0x2701A
nop
nop
ori         $r13, $r13, 0x125
nop
nop
```

```
lu12i.w     $r14, 0xE5B73
nop
nop
ori         $r14, $r14, 0x9BD
nop
nop
lu12i.w     $r15, 0xA8E89
nop
nop
ori         $r15, $r15, 0x1CD
nop
nop
lu12i.w     $r16, 0xD497
nop
nop
ori         $r16, $r16, 0x74C
nop
nop
lu12i.w     $r17, 0x188DB
nop
nop
ori         $r17, $r17, 0x25C
nop
nop
lu12i.w     $r18, 0x9EEAB
nop
nop
ori         $r18, $r18, 0xF1C
nop
nop
lu12i.w     $r19, 0xA6E7C
nop
nop
ori         $r19, $r19, 0x132
nop
nop
lu12i.w     $r20, 0x3093F
nop
nop
ori         $r20, $r20, 0x94C
nop
nop
lu12i.w     $r21, 0xF89EC
nop
nop
ori         $r21, $r21, 0xD69
nop
nop
lu12i.w     $r22, 0x9B2AF
nop
nop
ori         $r22, $r22, 0xA0D
nop
nop
```

```
lu12i.w     $r23, 0xB4F21
nop
nop
ori         $r23, $r23, 0x7ED
nop
nop
lu12i.w     $r24, 0xA1972
nop
nop
ori         $r24, $r24, 0x9DD
nop
nop
lu12i.w     $r25, 0xF2217
nop
nop
ori         $r25, $r25, 0x8F
nop
nop
lu12i.w     $r26, 0x4AEB7
nop
nop
ori         $r26, $r26, 0xA9A
nop
nop
lu12i.w     $r27, 0xBCE91
nop
nop
ori         $r27, $r27, 0x641
nop
nop
lu12i.w     $r28, 0x39D9E
nop
nop
ori         $r28, $r28, 0xB87
nop
nop
lu12i.w     $r29, 0xFBAB3
nop
nop
ori         $r29, $r29, 0xE8C
nop
nop
lu12i.w     $r30, 0xCC476
nop
nop
ori         $r30, $r30, 0x127
nop
nop
lu12i.w     $r31, 0x679D7
nop
nop
ori         $r31, $r31, 0xCD8
nop
nop
```

```
L1: # Test 0-0: sub.w
    sub.w       $r3, $r20, $r23
    nop
    nop
    lu12i.w     $r31, 0x7BA1E
    nop
    nop
    ori         $r31, $r31, 0x15F
    nop
    nop
    bne         $r3, $r31, fail
    nop
    nop
L2: # Test 0-1: sub.w
    sub.w       $r6, $r3, $r18
    nop
    nop
    lu12i.w     $r2, 0xDCB72
    nop
    nop
    ori         $r2, $r2, 0x243
    nop
    nop
    bne         $r6, $r2, fail
    nop
    nop
L3: # Test 1-0: sra.w
    sra.w       $r26, $r10, $r8
    nop
    nop
    lu12i.w     $r8, 0xFFFFF
    nop
    nop
    ori         $r8, $r8, 0xFF9
    nop
    nop
    bne         $r26, $r8, fail
    nop
    nop
L4: # Test 1-1: sra.w
    sra.w       $r5, $r10, $r21
    nop
    nop
    lu12i.w     $r2, 0xFFE75
    nop
    nop
    ori         $r2, $r2, 0x3B8
    nop
    nop
    bne         $r5, $r2, fail
    nop
    nop
L5: # Test 2-0: slt
    slt         $r19, $r30, $r18
```

```
    nop
    nop
    ori         $r26, $r0, 0x0
    nop
    nop
    bne         $r19, $r26, fail
    nop
    nop
L6: # Test 2-1: slt
    slt         $r27, $r28, $r11
    nop
    nop
    ori         $r11, $r0, 0x0
    nop
    nop
    bne         $r27, $r11, fail
    nop
    nop
L7: # Test 3-0: sltu
    sltu        $r3, $r25, $r17
    nop
    nop
    ori         $r20, $r0, 0x0
    nop
    nop
    bne         $r3, $r20, fail
    nop
    nop
L8: # Test 3-1: sltu
    sltu        $r25, $r4, $r15
    nop
    nop
    ori         $r6, $r0, 0x0
    nop
    nop
    bne         $r25, $r6, fail
    nop
    nop
L9: # Test 4-0: slli.w
    slli.w      $r30, $r20, 0xB
    nop
    nop
    ori         $r18, $r0, 0x0
    nop
    nop
    bne         $r30, $r18, fail
    nop
    nop
L10: # Test 4-1: slli.w
    slli.w      $r0, $r19, 0x19
    nop
    nop
    ori         $r4, $r0, 0x0
    nop
```

```
    nop
    bne         $r0, $r4, fail
    nop
    nop
L11: # Test 5-0: srli.w
    srli.w      $r14, $r23, 0x18
    nop
    nop
    ori         $r2, $r0, 0xB4
    nop
    nop
    bne         $r14, $r2, fail
    nop
    nop
L12: # Test 5-1: srli.w
    srli.w      $r11, $r19, 0x15
    nop
    nop
    ori         $r24, $r0, 0x0
    nop
    nop
    bne         $r11, $r24, fail
    nop
    nop
L13: # Test 6-0: slti
    slti        $r25, $r14, 0x735
    nop
    nop
    ori         $r30, $r0, 0x1
    nop
    nop
    bne         $r25, $r30, fail
    nop
    nop
L14: # Test 6-1: slti
    slti        $r1, $r24, -0x23E
    nop
    nop
    ori         $r28, $r0, 0x0
    nop
    nop
    bne         $r1, $r28, fail
    nop
    nop
L15: # Test 7-0: sltui
    sltui       $r15, $r4, 0x735
    nop
    nop
    ori         $r10, $r0, 0x1
    nop
    nop
    bne         $r15, $r10, fail
    nop
    nop
```

```
L16: # Test 7-1: sltui
    sltui       $r29, $r25, 0x4C5
    nop
    nop
    ori         $r22, $r0, 0x1
    nop
    nop
    bne         $r29, $r22, fail
    nop
    nop
L17: # Test 8-0: xori
    xori        $r13, $r0, 0x7AA
    nop
    nop
    ori         $r25, $r0, 0x7AA
    nop
    nop
    bne         $r13, $r25, fail
    nop
    nop
L18: # Test 8-1: xori
    xori        $r18, $r15, 0x1AC
    nop
    nop
    ori         $r7, $r0, 0x1AD
    nop
    nop
    bne         $r18, $r7, fail
    nop
    nop
L19: # Test 9-0: lu12i.w
    lu12i.w     $r7, 0x1939E
    nop
    nop
    lu12i.w     $r6, 0x1939E
    nop
    nop
    bne         $r7, $r6, fail
    nop
    nop
L20: # Test 9-1: lu12i.w
    lu12i.w     $r13, -0x4BCD3
    nop
    nop
    lu12i.w     $r12, 0xB432D
    nop
    nop
    bne         $r13, $r12, fail
    nop
    nop
L21: # Test 10-0: pcaddu12i
    pcaddu12i   $r31, 0x259D7
    nop
    nop
```

```
    lu12i.w     $r28, 0x419D7
    nop
    nop
    ori         $r28, $r28, 0x600
    nop
    nop
    bne         $r31, $r28, fail
    nop
    nop
L22: # Test 10-1: pcaddu12i
    pcaddu12i   $r28, -0x1AB42  #1c000600
    nop
    nop
    lu12i.w     $r4, 0x14BE
    nop
    nop
    ori         $r4, $r4, 0x630
    nop
    nop
    bne         $r28, $r4, fail
    nop
    nop
win: # Win label
    ori         $r4, $r0, 0x0
    nop
    nop
    halt
    nop
    nop
fail: # Fail label
    lu12i.w     $r4, 0xFFFFF
    nop
    nop
    ori         $r4, $r4, 0xFFF
    nop
    nop
    halt
    nop
    nop
```

测试程序 II

```
L0: # Randomly initialize GPR
    lu12i.w     $r0, 0x4DBCF
    nop
    nop
    ori         $r0, $r0, 0x432
    nop
    nop
    lu12i.w     $r1, 0x2445A
    nop
    nop
```

```
ori         $r1, $r1, 0xEA6
nop
nop
lu12i.w     $r2, 0x5FB20
nop
nop
ori         $r2, $r2, 0x8BC
nop
nop
lu12i.w     $r3, 0x75E8B
nop
nop
ori         $r3, $r3, 0x927
nop
nop
lu12i.w     $r4, 0x16A9
nop
nop
ori         $r4, $r4, 0x96F
nop
nop
lu12i.w     $r5, 0x8DEC1
nop
nop
ori         $r5, $r5, 0x8DA
nop
nop
lu12i.w     $r6, 0x495FE
nop
nop
ori         $r6, $r6, 0x9CB
nop
nop
lu12i.w     $r7, 0xE8B3E
nop
nop
ori         $r7, $r7, 0xDE8
nop
nop
lu12i.w     $r8, 0x8BE1E
nop
nop
ori         $r8, $r8, 0xFD0
nop
nop
lu12i.w     $r9, 0x3BCE3
nop
nop
ori         $r9, $r9, 0x2A1
nop
nop
lu12i.w     $r10, 0x83998
nop
nop
```

```
ori         $r10, $r10, 0xEA4
nop
nop
lu12i.w     $r11, 0x9F999
nop
nop
ori         $r11, $r11, 0x793
nop
nop
lu12i.w     $r12, 0x2E0FA
nop
nop
ori         $r12, $r12, 0x8A8
nop
nop
lu12i.w     $r13, 0xF2F30
nop
nop
ori         $r13, $r13, 0xD62
nop
nop
lu12i.w     $r14, 0x4019
nop
nop
ori         $r14, $r14, 0x80A
nop
nop
lu12i.w     $r15, 0x6109E
nop
nop
ori         $r15, $r15, 0xAD9
nop
nop
lu12i.w     $r16, 0xC676
nop
nop
ori         $r16, $r16, 0x8FB
nop
nop
lu12i.w     $r17, 0xAD8BA
nop
nop
ori         $r17, $r17, 0x1BD
nop
nop
lu12i.w     $r18, 0xDD3B7
nop
nop
ori         $r18, $r18, 0x3EB
nop
nop
lu12i.w     $r19, 0x83F8A
nop
nop
```

```
ori         $r19, $r19, 0x517
nop
nop
lu12i.w     $r20, 0xCB8CD
nop
nop
ori         $r20, $r20, 0xA5E
nop
nop
lu12i.w     $r21, 0x7FD17
nop
nop
ori         $r21, $r21, 0xBC1
nop
nop
lu12i.w     $r22, 0xA7E69
nop
nop
ori         $r22, $r22, 0xDE8
nop
nop
lu12i.w     $r23, 0x6B9E1
nop
nop
ori         $r23, $r23, 0x751
nop
nop
lu12i.w     $r24, 0xF2089
nop
nop
ori         $r24, $r24, 0xF91
nop
nop
lu12i.w     $r25, 0xCA329
nop
nop
ori         $r25, $r25, 0xFA7
nop
nop
lu12i.w     $r26, 0x6BC44
nop
nop
ori         $r26, $r26, 0xDC7
nop
nop
lu12i.w     $r27, 0x60CD1
nop
nop
ori         $r27, $r27, 0x34
nop
nop
lu12i.w     $r28, 0x7A04
nop
nop
```

```
    ori         $r28, $r28, 0xAFA
    nop
    nop
    lu12i.w     $r29, 0xD954
    nop
    nop
    ori         $r29, $r29, 0x809
    nop
    nop
    lu12i.w     $r30, 0x5FC66
    nop
    nop
    ori         $r30, $r30, 0x19E
    nop
    nop
    lu12i.w     $r31, 0xF6925
    nop
    nop
    ori         $r31, $r31, 0x31A
    nop
    nop
L1: # Test 0-0: ld.w
    lu12i.w     $r28, 0x1C800
    nop
    nop
    ori         $r28, $r28, 0xC67
    nop
    nop
    ld.w        $r4, $r28, 0x41D
    nop
    nop
    ori         $r8, $r0, 0x0
    nop
    nop
    bne         $r4, $r8, fail
    nop
    nop
L2: # Test 1-0: ld.h
    lu12i.w     $r18, 0x1C800
    nop
    nop
    ori         $r18, $r18, 0x745
    nop
    nop
    ld.h        $r21, $r18, 0x13B
    nop
    nop
    ori         $r13, $r0, 0x0
    nop
    nop
    bne         $r21, $r13, fail
    nop
    nop
L3: # Test 2-0: ld.bu
```

```
    lu12i.w     $r6, 0x1C800
    nop
    nop
    ori         $r6, $r6, 0x9D5
    nop
    nop
    ld.bu       $r12, $r6, -0x38F
    nop
    nop
    ori         $r23, $r0, 0x0
    nop
    nop
    bne         $r12, $r23, fail
    nop
    nop
L4: # Test 3-0: st.h
    lu12i.w     $r15, 0x1C802
    nop
    nop
    ori         $r15, $r15, 0xD6
    nop
    nop
    st.h        $r18, $r15, -0x2F4
    nop
    nop
    lu12i.w     $r12, 0x1C802
    nop
    nop
    ori         $r12, $r12, 0x17A
    nop
    nop
    ld.bu       $r13, $r12, -0x39A
    nop
    nop
    ori         $r18, $r0, 0x0
    nop
    nop
    bne         $r13, $r18, fail
    nop
    nop
    lu12i.w     $r31, 0x1C802
    nop
    nop
    ori         $r31, $r31, 0x2B6
    nop
    nop
    ld.bu       $r1, $r31, -0x4D6
    nop
    nop
    ori         $r30, $r0, 0x0
    nop
    nop
    bne         $r1, $r30, fail
    nop
```

```
    nop
    lu12i.w     $r9, 0x1C801
    nop
    nop
    ori         $r9, $r9, 0x999
    nop
    nop
    ld.bu       $r21, $r9, 0x447
    nop
    nop
    ori         $r16, $r0, 0x0
    nop
    nop
    bne         $r21, $r16, fail
    nop
    nop
    lu12i.w     $r19, 0x1C801
    nop
    nop
    ori         $r19, $r19, 0x74F
    nop
    nop
    ld.bu       $r4, $r19, 0x691
    nop
    nop
    ori         $r21, $r0, 0x0
    nop
    nop
    bne         $r4, $r21, fail
    nop
    nop
L5: # Test 4-0: st.b
    lu12i.w     $r14, 0x1C801
    nop
    nop
    ori         $r14, $r14, 0xC9C
    nop
    nop
    st.b        $r26, $r14, -0x72
    nop
    nop
    lu12i.w     $r7, 0x1C801
    nop
    nop
    ori         $r7, $r7, 0xEDF
    nop
    nop
    ld.bu       $r6, $r7, -0x2B7
    nop
    nop
    ori         $r12, $r0, 0x0
    nop
    nop
    bne         $r6, $r12, fail
```

```
    nop
    nop
    lu12i.w     $r3, 0x1C801
    nop
    nop
    ori         $r3, $r3, 0x686
    nop
    nop
    ld.bu       $r22, $r3, 0x5A2
    nop
    nop
    ori         $r23, $r0, 0x0
    nop
    nop
    bne         $r22, $r23, fail
    nop
    nop
    lu12i.w     $r30, 0x1C801
    nop
    nop
    ori         $r30, $r30, 0x64D
    nop
    nop
    ld.bu       $r14, $r30, 0x5DB
    nop
    nop
    ori         $r21, $r0, 0x0
    nop
    nop
    bne         $r14, $r21, fail
    nop
    nop
    lu12i.w     $r17, 0x1C801
    nop
    nop
    ori         $r17, $r17, 0x8B8
    nop
    nop
    ld.bu       $r7, $r17, 0x370
    nop
    nop
    ori         $r8, $r0, 0x0
    nop
    nop
    bne         $r7, $r8, fail
    nop
    nop
L6: # Test 5-0: b
    b           L7
    nop
    nop
    b           fail
    nop
    nop
```

```
L7: # Test 6-0: bl
    bl          L8
    nop
    nop
    b           fail
    nop
    nop
L8: # bl target
    lu12i.w     $r3, 0x1C000
    nop
    nop
    ori         $r3, $r3, 0x5F8
    nop
    nop
    bne         $r1, $r3, fail
    nop
    nop
L9: # Test 7-0: jirl
    lu12i.w     $r27, 0x1C013
    nop
    nop
    ori         $r27, $r27, 0x600
    nop
    nop
    jirl        $r1, $r27, -0x12FA0
    nop
    nop
    b           fail
    nop
    nop
L10: # jirl target
    lu12i.w     $r7, 0x1C000
    nop
    nop
    ori         $r7, $r7, 0x64C
    nop
    nop
    bne         $r1, $r7, fail
    nop
    nop
L11: # Test 8-0: beq
    beq         $r23, $r19, fail
    nop
    nop
L12: # Test 9-0: bne
    bne         $r13, $r21, fail
    nop
    nop
L13: # Test 10-0: blt
    blt         $r18, $r16, fail
    nop
    nop
L14: # Test 11-0: bge
    bge         $r4, $r9, fail
```

```
    nop
    nop
L15: # Test 12-0: bltu
    bltu        $r21, $r4, fail
    nop
    nop
L16: # Test 13-0: bgeu
    bgeu        $r1, $r21, win
    nop
    nop
    b           fail
    nop
    nop
win: # Win label
    ori         $r4, $r0, 0x0
    nop
    nop
    halt
    nop
    nop
fail: # Fail label
    lu12i.w     $r4, 0xFFFFF
    nop
    nop
    ori         $r4, $r4, 0xFFF
    nop
    nop
    halt
    nop
    nop
```

测试程序 III

```
L0: # Randomly initialize GPR
    lu12i.w     $r0, 0x3225B
    ori         $r0, $r0, 0xA15
    lu12i.w     $r1, 0xA5A95
    ori         $r1, $r1, 0x228
    lu12i.w     $r2, 0xAB0E1
    ori         $r2, $r2, 0x9C7
    lu12i.w     $r3, 0x2E189
    ori         $r3, $r3, 0xA07
    lu12i.w     $r4, 0x52C9C
    ori         $r4, $r4, 0xD3B
    lu12i.w     $r5, 0x49321
    ori         $r5, $r5, 0x365
    lu12i.w     $r6, 0x9C121
    ori         $r6, $r6, 0xCE8
    lu12i.w     $r7, 0x35252
    ori         $r7, $r7, 0x806
    lu12i.w     $r8, 0x7B874
    ori         $r8, $r8, 0x44E
```

```
    lu12i.w     $r9, 0xC16E1
    ori         $r9, $r9, 0x7C3
    lu12i.w     $r10, 0x800E4
    ori         $r10, $r10, 0xE03
    lu12i.w     $r11, 0xFA912
    ori         $r11, $r11, 0xCED
    lu12i.w     $r12, 0xFCACF
    ori         $r12, $r12, 0xE88
    lu12i.w     $r13, 0xACC37
    ori         $r13, $r13, 0x236
    lu12i.w     $r14, 0xE6C96
    ori         $r14, $r14, 0x9C2
    lu12i.w     $r15, 0x647D4
    ori         $r15, $r15, 0xCD8
    lu12i.w     $r16, 0x27E4D
    ori         $r16, $r16, 0xE0D
    lu12i.w     $r17, 0x1F25E
    ori         $r17, $r17, 0x80
    lu12i.w     $r18, 0xEC8F7
    ori         $r18, $r18, 0x296
    lu12i.w     $r19, 0xEE3D
    ori         $r19, $r19, 0x810
    lu12i.w     $r20, 0x43452
    ori         $r20, $r20, 0x9
    lu12i.w     $r21, 0x2F419
    ori         $r21, $r21, 0x4D0
    lu12i.w     $r22, 0xFE56A
    ori         $r22, $r22, 0x3B0
    lu12i.w     $r23, 0x5C787
    ori         $r23, $r23, 0x8F9
    lu12i.w     $r24, 0x7D279
    ori         $r24, $r24, 0x1DE
    lu12i.w     $r25, 0x4BB00
    ori         $r25, $r25, 0xC03
    lu12i.w     $r26, 0x3B8A3
    ori         $r26, $r26, 0x495
    lu12i.w     $r27, 0x7BC70
    ori         $r27, $r27, 0x9D8
    lu12i.w     $r28, 0x59F14
    ori         $r28, $r28, 0xE15
    lu12i.w     $r29, 0xDD55B
    ori         $r29, $r29, 0xA73
    lu12i.w     $r30, 0x671F2
    ori         $r30, $r30, 0xA19
    lu12i.w     $r31, 0xF094C
    ori         $r31, $r31, 0xAB2
L1: # Test 0-0: add.w
    add.w       $r20, $r18, $r5
    lu12i.w     $r11, 0x35C18
    ori         $r11, $r11, 0x5FB
    bne         $r20, $r11, fail
    nop
L2: # Test 0-1: add.w
    add.w       $r5, $r16, $r2
```

```
    lu12i.w     $r9, 0xD2F2F
    ori         $r9, $r9, 0x7D4
    bne         $r5, $r9, fail
    nop
L3: # Test 0-2: add.w
    add.w       $r27, $r2, $r2
    lu12i.w     $r20, 0x561C3
    ori         $r20, $r20, 0x38E
    bne         $r27, $r20, fail
    nop
L4: # Test 0-3: add.w
    add.w       $r22, $r9, $r2
    lu12i.w     $r25, 0x7E011
    ori         $r25, $r25, 0x19B
    bne         $r22, $r25, fail
    nop
L5: # Test 0-4: add.w
    add.w       $r4, $r12, $r24
    lu12i.w     $r28, 0x79D49
    ori         $r28, $r28, 0x66
    bne         $r4, $r28, fail
    nop
L6: # Test 1-0: sub.w
    sub.w       $r26, $r26, $r28
    lu12i.w     $r24, 0xC1B5A
    ori         $r24, $r24, 0x42F
    bne         $r26, $r24, fail
    nop
L7: # Test 1-1: sub.w
    sub.w       $r10, $r23, $r18
    lu12i.w     $r5, 0x6FE90
    ori         $r5, $r5, 0x663
    bne         $r10, $r5, fail
    nop
L8: # Test 1-2: sub.w
    sub.w       $r27, $r12, $r11
    lu12i.w     $r14, 0xC6EB7
    ori         $r14, $r14, 0x88D
    bne         $r27, $r14, fail
    nop
L9: # Test 1-3: sub.w
    sub.w       $r28, $r27, $r20
    lu12i.w     $r18, 0x70CF4
    ori         $r18, $r18, 0x4FF
    bne         $r28, $r18, fail
    nop
L10: # Test 1-4: sub.w
    sub.w       $r13, $r28, $r10
    lu12i.w     $r26, 0xE63
    ori         $r26, $r26, 0xE9C
    bne         $r13, $r26, fail
    nop
L11: # Test 2-0: sll.w
    sll.w       $r19, $r24, $r21
```

```
    lu12i.w     $r12, 0xA42F0
    bne         $r19, $r12, fail
    nop
L12: # Test 2-1: sll.w
    sll.w       $r15, $r14, $r10
    lu12i.w     $r12, 0x375BC
    ori         $r12, $r12, 0x468
    bne         $r15, $r12, fail
    nop
L13: # Test 2-2: sll.w
    sll.w       $r5, $r31, $r13
    lu12i.w     $r14, 0x20000
    bne         $r5, $r14, fail
    nop
L14: # Test 2-3: sll.w
    sll.w       $r26, $r11, $r12
    lu12i.w     $r6, 0xC185F
    ori         $r6, $r6, 0xB00
    bne         $r26, $r6, fail
    nop
L15: # Test 2-4: sll.w
    sll.w       $r29, $r11, $r28
    lu12i.w     $r25, 0x80000
    bne         $r29, $r25, fail
    nop
L16: # Test 3-0: srl.w
    srl.w       $r23, $r9, $r27
    lu12i.w     $r16, 0x69
    ori         $r16, $r16, 0x797
    bne         $r23, $r16, fail
    nop
L17: # Test 3-1: srl.w
    srl.w       $r8, $r17, $r9
    ori         $r27, $r0, 0x1F2
    bne         $r8, $r27, fail
    nop
L18: # Test 3-2: srl.w
    srl.w       $r4, $r14, $r1
    lu12i.w     $r6, 0x200
    bne         $r4, $r6, fail
    nop
L19: # Test 3-3: srl.w
    srl.w       $r15, $r14, $r21
    lu12i.w     $r21, 0x2
    bne         $r15, $r21, fail
    nop
L20: # Test 3-4: srl.w
    srl.w       $r15, $r9, $r7
    lu12i.w     $r22, 0x34BC
    ori         $r22, $r22, 0xBDF
    bne         $r15, $r22, fail
    nop
L21: # Test 4-0: sra.w
    sra.w       $r13, $r6, $r23
```

```
    ori         $r14, $r0, 0x0
    bne         $r13, $r14, fail
    nop
L22: # Test 4-1: sra.w
    sra.w       $r12, $r13, $r30
    ori         $r11, $r0, 0x0
    bne         $r12, $r11, fail
    nop
L23: # Test 4-2: sra.w
    sra.w       $r18, $r23, $r3
    ori         $r4, $r0, 0xD2F
    bne         $r18, $r4, fail
    nop
L24: # Test 4-3: sra.w
    sra.w       $r3, $r21, $r1
    ori         $r4, $r0, 0x20
    bne         $r3, $r4, fail
    nop
L25: # Test 4-4: sra.w
    sra.w       $r20, $r11, $r15
    ori         $r3, $r0, 0x0
    bne         $r20, $r3, fail
    nop
L26: # Test 5-0: sltu
    sltu        $r18, $r8, $r13
    ori         $r14, $r0, 0x0
    bne         $r18, $r14, fail
    nop
L27: # Test 5-1: sltu
    sltu        $r25, $r26, $r1
    ori         $r29, $r0, 0x0
    bne         $r25, $r29, fail
    nop
L28: # Test 5-2: sltu
    sltu        $r31, $r7, $r23
    ori         $r26, $r0, 0x0
    bne         $r31, $r26, fail
    nop
L29: # Test 5-3: sltu
    sltu        $r14, $r23, $r13
    ori         $r13, $r0, 0x0
    bne         $r14, $r13, fail
    nop
L30: # Test 5-4: sltu
    sltu        $r11, $r16, $r27
    ori         $r19, $r0, 0x0
    bne         $r11, $r19, fail
    nop
L31: # Test 6-0: or
    or          $r2, $r5, $r17
    lu12i.w     $r14, 0x3F25E
    ori         $r14, $r14, 0x80
    bne         $r2, $r14, fail
    nop
```

```
L32: # Test 6-1: or
    or          $r0, $r0, $r16
    ori         $r18, $r0, 0x0
    bne         $r0, $r18, fail
    nop
L33: # Test 6-2: or
    or          $r21, $r13, $r1
    lu12i.w     $r15, 0xA5A95
    ori         $r15, $r15, 0x228
    bne         $r21, $r15, fail
    nop
L34: # Test 6-3: or
    or          $r25, $r19, $r8
    ori         $r30, $r0, 0x1F2
    bne         $r25, $r30, fail
    nop
L35: # Test 6-4: or
    or          $r22, $r15, $r15
    lu12i.w     $r28, 0xA5A95
    ori         $r28, $r28, 0x228
    bne         $r22, $r28, fail
    nop
L36: # Test 7-0: xor
    xor         $r14, $r25, $r29
    ori         $r26, $r0, 0x1F2
    bne         $r14, $r26, fail
    nop
L37: # Test 7-1: xor
    xor         $r13, $r4, $r14
    ori         $r25, $r0, 0x1D2
    bne         $r13, $r25, fail
    nop
L38: # Test 7-2: xor
    xor         $r30, $r0, $r7
    lu12i.w     $r4, 0x35252
    ori         $r4, $r4, 0x806
    bne         $r30, $r4, fail
    nop
L39: # Test 7-3: xor
    xor         $r28, $r29, $r5
    lu12i.w     $r15, 0x20000
    bne         $r28, $r15, fail
    nop
L40: # Test 7-4: xor
    xor         $r26, $r19, $r17
    lu12i.w     $r1, 0x1F25E
    ori         $r1, $r1, 0x80
    bne         $r26, $r1, fail
    nop
L41: # Test 8-0: addi.w
    addi.w      $r22, $r30, -0xC9
    lu12i.w     $r7, 0x35252
    ori         $r7, $r7, 0x73D
    bne         $r22, $r7, fail
```

```
    nop
L42: # Test 8-1: addi.w
    addi.w      $r18, $r15, -0x2C8
    lu12i.w     $r2, 0x1FFFF
    ori         $r2, $r2, 0xD38
    bne         $r18, $r2, fail
    nop
L43: # Test 8-2: addi.w
    addi.w      $r19, $r22, 0x4B7
    lu12i.w     $r7, 0x35252
    ori         $r7, $r7, 0xBF4
    bne         $r19, $r7, fail
    nop
L44: # Test 8-3: addi.w
    addi.w      $r29, $r15, 0x204
    lu12i.w     $r31, 0x20000
    ori         $r31, $r31, 0x204
    bne         $r29, $r31, fail
    nop
L45: # Test 8-4: addi.w
    addi.w      $r26, $r14, 0x6F4
    ori         $r1, $r0, 0x8E6
    bne         $r26, $r1, fail
    nop
L46: # Test 9-0: slli.w
    slli.w      $r10, $r3, 0xD
    ori         $r11, $r0, 0x0
    bne         $r10, $r11, fail
    nop
L47: # Test 9-1: slli.w
    slli.w      $r17, $r12, 0x6
    ori         $r4, $r0, 0x0
    bne         $r17, $r4, fail
    nop
L48: # Test 9-2: slli.w
    slli.w      $r4, $r26, 0x15
    lu12i.w     $r14, 0x1CC00
    bne         $r4, $r14, fail
    nop
L49: # Test 9-3: slli.w
    slli.w      $r26, $r27, 0x15
    lu12i.w     $r17, 0x3E400
    bne         $r26, $r17, fail
    nop
L50: # Test 9-4: slli.w
    slli.w      $r18, $r20, 0xA
    ori         $r12, $r0, 0x0
    bne         $r18, $r12, fail
    nop
L51: # Test 10-0: srai.w
    srai.w      $r4, $r4, 0x14
    ori         $r8, $r0, 0x1CC
    bne         $r4, $r8, fail
    nop
```

```
L52: # Test 10-1: srai.w
    srai.w      $r5, $r28, 0x1C
    ori         $r4, $r0, 0x2
    bne         $r5, $r4, fail
    nop
L53: # Test 10-2: srai.w
    srai.w      $r14, $r11, 0xE
    ori         $r17, $r0, 0x0
    bne         $r14, $r17, fail
    nop
L54: # Test 10-3: srai.w
    srai.w      $r7, $r7, 0x7
    lu12i.w     $r9, 0x6A4
    ori         $r9, $r9, 0xA57
    bne         $r7, $r9, fail
    nop
L55: # Test 10-4: srai.w
    srai.w      $r12, $r31, 0x1A
    ori         $r9, $r0, 0x8
    bne         $r12, $r9, fail
    nop
L56: # Test 11-0: slti
    slti        $r22, $r19, -0x3C1
    ori         $r5, $r0, 0x0
    bne         $r22, $r5, fail
    nop
L57: # Test 11-1: slti
    slti        $r8, $r31, -0x7B4
    ori         $r11, $r0, 0x0
    bne         $r8, $r11, fail
    nop
L58: # Test 11-2: slti
    slti        $r17, $r5, -0x37B
    ori         $r1, $r0, 0x0
    bne         $r17, $r1, fail
    nop
L59: # Test 11-3: slti
    slti        $r9, $r18, -0x126
    ori         $r20, $r0, 0x0
    bne         $r9, $r20, fail
    nop
L60: # Test 11-4: slti
    slti        $r14, $r5, 0x144
    ori         $r31, $r0, 0x1
    bne         $r14, $r31, fail
    nop
L61: # Test 12-0: sltui
    sltui       $r4, $r2, -0x312
    ori         $r1, $r0, 0x1
    bne         $r4, $r1, fail
    nop
L62: # Test 12-1: sltui
    sltui       $r1, $r17, -0x52E
    ori         $r27, $r0, 0x1
```

```
    bne         $r1, $r27, fail
    nop
L63: # Test 12-2: sltui
    sltui       $r27, $r27, 0x2A1
    ori         $r15, $r0, 0x1
    bne         $r27, $r15, fail
    nop
L64: # Test 12-3: sltui
    sltui       $r19, $r12, 0x2CE
    ori         $r10, $r0, 0x1
    bne         $r19, $r10, fail
    nop
L65: # Test 12-4: sltui
    sltui       $r25, $r4, -0x782
    ori         $r5, $r0, 0x1
    bne         $r25, $r5, fail
    nop
L66: # Test 13-0: andi
    andi        $r7, $r29, 0xF9C
    ori         $r31, $r0, 0x204
    bne         $r7, $r31, fail
    nop
L67: # Test 13-1: andi
    andi        $r18, $r29, 0x76
    ori         $r16, $r0, 0x4
    bne         $r18, $r16, fail
    nop
L68: # Test 13-2: andi
    andi        $r27, $r3, 0x7A8
    ori         $r5, $r0, 0x0
    bne         $r27, $r5, fail
    nop
L69: # Test 13-3: andi
    andi        $r18, $r19, 0x207
    ori         $r30, $r0, 0x1
    bne         $r18, $r30, fail
    nop
L70: # Test 13-4: andi
    andi        $r0, $r1, 0x7C1
    ori         $r6, $r0, 0x0
    bne         $r0, $r6, fail
    nop
L71: # Test 14-0: ori
    ori         $r3, $r2, 0x102
    lu12i.w     $r24, 0x1FFFF
    ori         $r24, $r24, 0xD3A
    bne         $r3, $r24, fail
    nop
L72: # Test 14-1: ori
    ori         $r28, $r13, 0x52B
    ori         $r30, $r0, 0x5FB
    bne         $r28, $r30, fail
    nop
L73: # Test 14-2: ori
```

```
    ori         $r15, $r16, 0x907
    ori         $r6, $r0, 0x907
    bne         $r15, $r6, fail
    nop
L74: # Test 14-3: ori
    ori         $r24, $r29, 0x106
    lu12i.w     $r31, 0x20000
    ori         $r31, $r31, 0x306
    bne         $r24, $r31, fail
    nop
L75: # Test 14-4: ori
    ori         $r2, $r16, 0x202
    ori         $r20, $r0, 0x206
    bne         $r2, $r20, fail
    nop
L76: # Test 15-0: lu12i.w
    lu12i.w     $r13, 0x2DDC4
    lu12i.w     $r10, 0x2DDC4
    bne         $r13, $r10, fail
    nop
L77: # Test 15-1: lu12i.w
    lu12i.w     $r2, 0x45A2F
    lu12i.w     $r20, 0x45A2F
    bne         $r2, $r20, fail
    nop
L78: # Test 15-2: lu12i.w
    lu12i.w     $r21, -0x65E01
    lu12i.w     $r16, 0x9A1FF
    bne         $r21, $r16, fail
    nop
L79: # Test 15-3: lu12i.w
    lu12i.w     $r31, -0x33E0A
    lu12i.w     $r15, 0xCC1F6
    bne         $r31, $r15, fail
    nop
L80: # Test 15-4: lu12i.w
    lu12i.w     $r28, -0x496AC
    lu12i.w     $r31, 0xB6954
    bne         $r28, $r31, fail
    nop
L81: # Test 16-0: pcaddu12i
    pcaddu12i   $r30, 0x69979
    lu12i.w     $r18, 0x85979
    ori         $r18, $r18, 0x668
    bne         $r30, $r18, fail
    nop
L82: # Test 16-1: pcaddu12i
    pcaddu12i   $r13, -0x70E06
    lu12i.w     $r18, 0xAB1FA
    ori         $r18, $r18, 0x67C
    bne         $r13, $r18, fail
    nop
L83: # Test 16-2: pcaddu12i
    pcaddu12i   $r7, -0x670C2
```

```
    lu12i.w     $r20, 0xB4F3E
    ori         $r20, $r20, 0x690
    bne         $r7, $r20, fail
    nop
L84: # Test 16-3: pcaddu12i
    pcaddu12i   $r7, -0x43EE7
    lu12i.w     $r4, 0xD8119
    ori         $r4, $r4, 0x6A4
    bne         $r7, $r4, fail
    nop
L85: # Test 16-4: pcaddu12i
    pcaddu12i   $r4, 0x7FE91
    lu12i.w     $r7, 0x9BE91
    ori         $r7, $r7, 0x6B8
    bne         $r4, $r7, fail
    nop
win: # Win label
    ori         $r4, $r0, 0x0
    halt
    nop
    nop
    nop
    nop
fail: # Fail label
    lu12i.w     $r4, 0xFFFFF
    ori         $r4, $r4, 0xFFF
    halt
```

测试程序 IV

```
L0: # Randomly initialize GPR
    lu12i.w     $r0, 0x465A7
    ori         $r0, $r0, 0x2ED
    lu12i.w     $r1, 0x4C8AC
    ori         $r1, $r1, 0x28A
    lu12i.w     $r2, 0x713D6
    ori         $r2, $r2, 0xEDA
    lu12i.w     $r3, 0xE7445
    ori         $r3, $r3, 0x26E
    lu12i.w     $r4, 0x7999E
    ori         $r4, $r4, 0x477
    lu12i.w     $r5, 0xC70C4
    ori         $r5, $r5, 0xA7E
    lu12i.w     $r6, 0xED86C
    ori         $r6, $r6, 0x9E4
    lu12i.w     $r7, 0xD9FD1
    ori         $r7, $r7, 0x76A
    lu12i.w     $r8, 0xB14CA
    ori         $r8, $r8, 0x60F
    lu12i.w     $r9, 0xBE487
    ori         $r9, $r9, 0x436
    lu12i.w     $r10, 0x590A1
```

```
    ori         $r10, $r10, 0x92C
    lu12i.w     $r11, 0x54009
    ori         $r11, $r11, 0x525
    lu12i.w     $r12, 0xE4676
    ori         $r12, $r12, 0xFB1
    lu12i.w     $r13, 0x83BDF
    ori         $r13, $r13, 0x952
    lu12i.w     $r14, 0xEDFB2
    ori         $r14, $r14, 0xC66
    lu12i.w     $r15, 0x66CC0
    ori         $r15, $r15, 0xDAB
    lu12i.w     $r16, 0x6F2C5
    ori         $r16, $r16, 0xA21
    lu12i.w     $r17, 0xD0D8A
    ori         $r17, $r17, 0xA03
    lu12i.w     $r18, 0x3FE6F
    ori         $r18, $r18, 0xEF6
    lu12i.w     $r19, 0xBE4A9
    ori         $r19, $r19, 0x7F9
    lu12i.w     $r20, 0x8BAFD
    ori         $r20, $r20, 0xDA7
    lu12i.w     $r21, 0xD6FE4
    ori         $r21, $r21, 0x832
    lu12i.w     $r22, 0x6F19E
    ori         $r22, $r22, 0xE7A
    lu12i.w     $r23, 0x5643
    ori         $r23, $r23, 0x5B8
    lu12i.w     $r24, 0xDC5E5
    ori         $r24, $r24, 0xA1B
    lu12i.w     $r25, 0xC6477
    ori         $r25, $r25, 0x18
    lu12i.w     $r26, 0xD887
    ori         $r26, $r26, 0x7AD
    lu12i.w     $r27, 0x20176
    ori         $r27, $r27, 0xE80
    lu12i.w     $r28, 0xC655D
    ori         $r28, $r28, 0xBB7
    lu12i.w     $r29, 0xF652A
    ori         $r29, $r29, 0xBF3
    lu12i.w     $r30, 0x66B73
    ori         $r30, $r30, 0xCC0
    lu12i.w     $r31, 0xAAD39
    ori         $r31, $r31, 0x336
L1: # Test 0-0: ld.h
    lu12i.w     $r9, 0x1C801
    ori         $r9, $r9, 0x5AF
    ld.h        $r8, $r9, -0x3CB
    ori         $r5, $r0, 0x0
    bne         $r8, $r5, fail
    b           L2
L2: # Test 0-1: ld.h
    lu12i.w     $r20, 0x1C802
    ori         $r20, $r20, 0x2B1
    ld.h        $r3, $r20, -0x28B
```

```
    ori         $r10, $r0, 0x0
    bne         $r3, $r10, fail
    b           L3
L3: # Test 0-2: ld.h
    lu12i.w     $r16, 0x1C801
    ori         $r16, $r16, 0x369
    ld.h        $r8, $r16, 0x317
    ori         $r20, $r0, 0x0
    bne         $r8, $r20, fail
    b           L4
L4: # Test 0-3: ld.h
    lu12i.w     $r12, 0x1C801
    ori         $r12, $r12, 0x1AF
    ld.h        $r4, $r12, -0x309
    ori         $r27, $r0, 0x0
    bne         $r4, $r27, fail
    b           L5
L5: # Test 1-0: ld.b
    lu12i.w     $r17, 0x1C7FF
    ori         $r17, $r17, 0xA94
    ld.b        $r10, $r17, 0x731
    ori         $r2, $r0, 0x0
    bne         $r10, $r2, fail
    b           L6
L6: # Test 1-1: ld.b
    lu12i.w     $r5, 0x1C802
    ori         $r5, $r5, 0x164
    ld.b        $r16, $r5, -0x3D5
    ori         $r6, $r0, 0x0
    bne         $r16, $r6, fail
    b           L7
L7: # Test 1-2: ld.b
    lu12i.w     $r17, 0x1C800
    ori         $r17, $r17, 0x4B
    ld.b        $r19, $r17, 0x1FF
    ori         $r5, $r0, 0x0
    bne         $r19, $r5, fail
    b           L8
L8: # Test 1-3: ld.b
    lu12i.w     $r23, 0x1C802
    ori         $r23, $r23, 0x9D3
    ld.b        $r30, $r23, -0x365
    ori         $r29, $r0, 0x0
    bne         $r30, $r29, fail
    b           L9
L9: # Test 2-0: ld.hu
    lu12i.w     $r18, 0x1C801
    ori         $r18, $r18, 0xBA
    ld.hu       $r14, $r18, -0x45E
    ori         $r8, $r0, 0x0
    bne         $r14, $r8, fail
    b           L10
L10: # Test 2-1: ld.hu
    lu12i.w     $r19, 0x1C800
```

```
    ori         $r19, $r19, 0xAF6
    ld.hu       $r10, $r19, -0xB2
    ori         $r30, $r0, 0x0
    bne         $r10, $r30, fail
    b           L11
L11: # Test 2-2: ld.hu
    lu12i.w     $r20, 0x1C801
    ori         $r20, $r20, 0x951
    ld.hu       $r3, $r20, -0x26B
    ori         $r11, $r0, 0x0
    bne         $r3, $r11, fail
    b           L12
L12: # Test 2-3: ld.hu
    lu12i.w     $r29, 0x1C801
    ori         $r29, $r29, 0x708
    ld.hu       $r29, $r29, 0x140
    ori         $r6, $r0, 0x0
    bne         $r29, $r6, fail
    b           L13
L13: # Test 3-0: st.w
    lu12i.w     $r3, 0x1C800
    ori         $r3, $r3, 0x4AF
    st.w        $r4, $r3, -0x333
    lu12i.w     $r30, 0x1C7FF
    ori         $r30, $r30, 0xB5F
    ld.bu       $r8, $r30, 0x61D
    ori         $r23, $r0, 0x0
    bne         $r8, $r23, fail
    lu12i.w     $r19, 0x1C800
    ori         $r19, $r19, 0x518
    ld.bu       $r18, $r19, -0x39C
    ori         $r7, $r0, 0x0
    bne         $r18, $r7, fail
    lu12i.w     $r27, 0x1C7FF
    ori         $r27, $r27, 0x9CF
    ld.bu       $r0, $r27, 0x7AD
    ori         $r12, $r0, 0x0
    bne         $r0, $r12, fail
    lu12i.w     $r29, 0x1C7FF
    ori         $r29, $r29, 0xDE0
    ld.bu       $r25, $r29, 0x39C
    ori         $r1, $r0, 0x0
    bne         $r25, $r1, fail
    b           L14
L14: # Test 3-1: st.w
    lu12i.w     $r23, 0x1C800
    ori         $r23, $r23, 0x917
    st.w        $r26, $r23, 0x1F9
    lu12i.w     $r6, 0x1C801
    ori         $r6, $r6, 0x81
    ld.bu       $r2, $r6, -0x571
    ori         $r30, $r0, 0xAD
    bne         $r2, $r30, fail
    lu12i.w     $r25, 0x1C800
```

```
    ori         $r25, $r25, 0x752
    ld.bu       $r8, $r25, 0x3BE
    ori         $r2, $r0, 0xAD
    bne         $r8, $r2, fail
    lu12i.w     $r12, 0x1C800
    ori         $r12, $r12, 0xFD6
    ld.bu       $r22, $r12, -0x4C6
    ori         $r25, $r0, 0xAD
    bne         $r22, $r25, fail
    lu12i.w     $r31, 0x1C800
    ori         $r31, $r31, 0x706
    ld.bu       $r21, $r31, 0x40A
    ori         $r14, $r0, 0xAD
    bne         $r21, $r14, fail
    b           L15
L15: # Test 3-2: st.w
    lu12i.w     $r26, 0x1C802
    ori         $r26, $r26, 0x39B
    st.w        $r18, $r26, 0x7DD
    lu12i.w     $r19, 0x1C802
    ori         $r19, $r19, 0xF2B
    ld.bu       $r25, $r19, -0x3B3
    ori         $r22, $r0, 0x0
    bne         $r25, $r22, fail
    lu12i.w     $r2, 0x1C802
    ori         $r2, $r2, 0xA82
    ld.bu       $r23, $r2, 0xF6
    ori         $r15, $r0, 0x0
    bne         $r23, $r15, fail
    lu12i.w     $r27, 0x1C802
    ori         $r27, $r27, 0x533
    ld.bu       $r27, $r27, 0x645
    ori         $r5, $r0, 0x0
    bne         $r27, $r5, fail
    lu12i.w     $r22, 0x1C802
    ori         $r22, $r22, 0x74A
    ld.bu       $r23, $r22, 0x42E
    ori         $r29, $r0, 0x0
    bne         $r23, $r29, fail
    b           L16
L16: # Test 3-3: st.w
    lu12i.w     $r8, 0x1C803
    ori         $r8, $r8, 0x2F0
    st.w        $r16, $r8, -0x73C
    lu12i.w     $r1, 0x1C802
    ori         $r1, $r1, 0x9B9
    ld.bu       $r18, $r1, 0x1FB
    ori         $r15, $r0, 0x0
    bne         $r18, $r15, fail
    lu12i.w     $r18, 0x1C802
    ori         $r18, $r18, 0xD8F
    ld.bu       $r19, $r18, -0x1DB
    ori         $r29, $r0, 0x0
    bne         $r19, $r29, fail
```

```
    lu12i.w     $r23, 0x1C803
    ori         $r23, $r23, 0x267
    ld.bu       $r26, $r23, -0x6B3
    ori         $r4, $r0, 0x0
    bne         $r26, $r4, fail
    lu12i.w     $r26, 0x1C802
    ori         $r26, $r26, 0x6BE
    ld.bu       $r10, $r26, 0x4F6
    ori         $r22, $r0, 0x0
    bne         $r10, $r22, fail
    b           L17
L17: # Test 4-0: st.h
    lu12i.w     $r16, 0x1C801
    ori         $r16, $r16, 0x143
    st.h        $r22, $r16, -0x59F
    lu12i.w     $r26, 0x1C801
    ori         $r26, $r26, 0x217
    ld.bu       $r10, $r26, -0x673
    ori         $r18, $r0, 0x0
    bne         $r10, $r18, fail
    lu12i.w     $r3, 0x1C800
    ori         $r3, $r3, 0x3F6
    ld.bu       $r7, $r3, 0x7AE
    ori         $r2, $r0, 0x0
    bne         $r7, $r2, fail
    lu12i.w     $r6, 0x1C800
    ori         $r6, $r6, 0xD49
    ld.bu       $r26, $r6, -0x1A5
    ori         $r24, $r0, 0x0
    bne         $r26, $r24, fail
    lu12i.w     $r15, 0x1C800
    ori         $r15, $r15, 0x605
    ld.bu       $r27, $r15, 0x59F
    ori         $r13, $r0, 0x0
    bne         $r27, $r13, fail
    b           L18
L18: # Test 4-1: st.h
    lu12i.w     $r9, 0x1C7FF
    ori         $r9, $r9, 0xF8A
    st.h        $r16, $r9, 0x3E8
    lu12i.w     $r9, 0x1C800
    ori         $r9, $r9, 0x460
    ld.bu       $r8, $r9, -0xF0
    ori         $r9, $r0, 0x0
    bne         $r8, $r9, fail
    lu12i.w     $r8, 0x1C800
    ori         $r8, $r8, 0x4BF
    ld.bu       $r9, $r8, -0x14F
    ori         $r3, $r0, 0x0
    bne         $r9, $r3, fail
    lu12i.w     $r10, 0x1C800
    ori         $r10, $r10, 0x267
    ld.bu       $r0, $r10, 0x109
    ori         $r17, $r0, 0x0
```

```
    bne         $r0, $r17, fail
    lu12i.w     $r19, 0x1C800
    ld.bu       $r9, $r19, 0x370
    ori         $r23, $r0, 0x0
    bne         $r9, $r23, fail
    b           L19
L19: # Test 4-2: st.h
    lu12i.w     $r2, 0x1C802
    ori         $r2, $r2, 0x6B7
    st.h        $r0, $r2, -0x335
    lu12i.w     $r7, 0x1C802
    ori         $r7, $r7, 0x2F3
    ld.bu       $r18, $r7, 0x8D
    ori         $r16, $r0, 0x0
    bne         $r18, $r16, fail
    lu12i.w     $r11, 0x1C801
    ori         $r11, $r11, 0xFCD
    ld.bu       $r4, $r11, 0x3B3
    ori         $r30, $r0, 0x0
    bne         $r4, $r30, fail
    lu12i.w     $r5, 0x1C802
    ori         $r5, $r5, 0x3C3
    ld.bu       $r2, $r5, -0x43
    ori         $r15, $r0, 0x0
    bne         $r2, $r15, fail
    lu12i.w     $r16, 0x1C802
    ori         $r16, $r16, 0x3E1
    ld.bu       $r0, $r16, -0x61
    ori         $r9, $r0, 0x0
    bne         $r0, $r9, fail
    b           L20
L20: # Test 4-3: st.h
    lu12i.w     $r3, 0x1C802
    ori         $r3, $r3, 0x9E6
    st.h        $r9, $r3, -0x766
    lu12i.w     $r9, 0x1C802
    ori         $r9, $r9, 0x4C9
    ld.bu       $r1, $r9, -0x249
    ori         $r11, $r0, 0x0
    bne         $r1, $r11, fail
    lu12i.w     $r9, 0x1C801
    ori         $r9, $r9, 0xE9B
    ld.bu       $r23, $r9, 0x3E5
    ori         $r9, $r0, 0x0
    bne         $r23, $r9, fail
    lu12i.w     $r16, 0x1C801
    ori         $r16, $r16, 0xE7C
    ld.bu       $r24, $r16, 0x404
    ori         $r17, $r0, 0x0
    bne         $r24, $r17, fail
    lu12i.w     $r16, 0x1C801
    ori         $r16, $r16, 0xC6E
    ld.bu       $r29, $r16, 0x612
    ori         $r4, $r0, 0x0
```

```
    bne         $r29, $r4, fail
    b           L21
L21: # Test 5-0: b
    b           L22
    b           fail
L22: # Test 5-1: b
    b           L23
    b           fail
L23: # Test 5-2: b
    b           L24
    b           fail
L24: # Test 5-3: b
    b           L25
    b           fail
L25: # Test 5-4: b
    b           L26
    b           fail
L26: # Test 6-0: bl
    bl          L27
    b           fail
L27: # bl target
    lu12i.w     $r14, 0x1C000
    ori         $r14, $r14, 0x548
    bne         $r1, $r14, fail
    b           L28
L28: # Test 6-1: bl
    bl          L29
    b           fail
L29: # bl target
    lu12i.w     $r28, 0x1C000
    ori         $r28, $r28, 0x560
    bne         $r1, $r28, fail
    b           L30
L30: # Test 6-2: bl
    bl          L31
    b           fail
L31: # bl target
    lu12i.w     $r15, 0x1C000
    ori         $r15, $r15, 0x578
    bne         $r1, $r15, fail
    b           L32
L32: # Test 6-3: bl
    bl          L33
    b           fail
L33: # bl target
    lu12i.w     $r10, 0x1C000
    ori         $r10, $r10, 0x590
    bne         $r1, $r10, fail
    b           L34
L34: # Test 6-4: bl
    bl          L35
    b           fail
L35: # bl target
    lu12i.w     $r15, 0x1C000
```

```
    ori         $r15, $r15, 0x5A8
    bne         $r1, $r15, fail
    b           L36
L36: # Test 7-0: jirl
    lu12i.w     $r8, 0x1BFF7
    ori         $r8, $r8, 0xA8C
    jirl        $r14, $r8, 0x8B40
    b           fail
L37: # jirl target
    lu12i.w     $r26, 0x1C000
    ori         $r26, $r26, 0x5C8
    bne         $r14, $r26, fail
    b           L38
L38: # Test 7-1: jirl
    lu12i.w     $r12, 0x1BFF7
    ori         $r12, $r12, 0x950
    jirl        $r29, $r12, 0x8C9C
    b           fail
L39: # jirl target
    lu12i.w     $r21, 0x1C000
    ori         $r21, $r21, 0x5E8
    bne         $r29, $r21, fail
    b           L40
L40: # Test 7-2: jirl
    lu12i.w     $r13, 0x1BFE0
    ori         $r13, $r13, 0x934
    jirl        $r22, $r13, 0x1FCD8
    b           fail
L41: # jirl target
    lu12i.w     $r15, 0x1C000
    ori         $r15, $r15, 0x608
    bne         $r22, $r15, fail
    b           L42
L42: # Test 7-3: jirl
    lu12i.w     $r2, 0x1BFFC
    ori         $r2, $r2, 0x868
    jirl        $r29, $r2, 0x3DC4
    b           fail
L43: # jirl target
    lu12i.w     $r8, 0x1C000
    ori         $r8, $r8, 0x628
    bne         $r29, $r8, fail
    b           L44
L44: # Test 7-4: jirl
    lu12i.w     $r3, 0x1C001
    ori         $r3, $r3, 0x244
    jirl        $r19, $r3, -0xBF8
    b           fail
L45: # jirl target
    lu12i.w     $r15, 0x1C000
    ori         $r15, $r15, 0x648
    bne         $r19, $r15, fail
    b           L46
L46: # Test 8-0: blt
```

```
    blt         $r15, $r14, fail
    b           L47
L47: # Test 8-1: blt
    blt         $r22, $r0, fail
    b           L48
L48: # Test 8-2: blt
    blt         $r10, $r8, L49
    b           fail
L49: # Test 8-3: blt
    blt         $r4, $r27, fail
    b           L50
L50: # Test 8-4: blt
    blt         $r13, $r21, L51
    b           fail
L51: # Test 9-0: bge
    bge         $r0, $r1, fail
    b           L52
L52: # Test 9-1: bge
    bge         $r15, $r24, L53
    b           fail
L53: # Test 9-2: bge
    bge         $r14, $r25, L54
    b           fail
L54: # Test 9-3: bge
    bge         $r3, $r8, L55
    b           fail
L55: # Test 9-4: bge
    bge         $r14, $r23, L56
    b           fail
L56: # Test 10-0: bltu
    bltu        $r4, $r25, fail
    b           L57
L57: # Test 10-1: bltu
    bltu        $r23, $r22, L58
    b           fail
L58: # Test 10-2: bltu
    bltu        $r14, $r27, fail
    b           L59
L59: # Test 10-3: bltu
    bltu        $r1, $r21, L60
    b           fail
L60: # Test 10-4: bltu
    bltu        $r2, $r13, fail
    b           L61
L61: # Test 11-0: bgeu
    bgeu        $r10, $r16, fail
    b           L62
L62: # Test 11-1: bgeu
    bgeu        $r15, $r12, L63
    b           fail
L63: # Test 11-2: bgeu
    bgeu        $r6, $r11, L64
    b           fail
L64: # Test 11-3: bgeu
```

```
    bgeu        $r26, $r4, L65
    b           fail
L65: # Test 11-4: bgeu
    bgeu        $r6, $r0, win
    b           fail
win: # Win label
    ori         $r4, $r0, 0x0
    halt
fail: # Fail label
    lu12i.w     $r4, 0xFFFFF
    ori         $r4, $r4, 0xFFF
    halt
```